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A Pinger Localization Algorithm Using Sparse Representation for Autonomous Underwater Vehicles

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***Abstract*—The problem of underwater Pinger localization is a crucial part of the underwater pipeline monitoring and maintenance by using an AUV in oil and gas industries. In this problem, a Pinger repeatedly transmits pulses of audio signals to an array of hydrophones. Due to the delay pattern, the direction of a Pinger relative to the hydrophone array can be estimated. However, since underwater environment is multipath in nature, multiple copies of Pinger pulses arrive at the hydrophones. In this paper, we employed the sparse representation to find the best combinations of the multipath signals, and use the direction of the first arrival path to be the direction of a Pinger.**

I. Introduction

The problem of underwater Pinger localization is a crucial part of the underwater pipeline monitoring and maintenance by using an AUV in oil and gas industries. In this problem, a Pinger repeatedly transmits pulses of audio signals to an array of hydrophones. The goal is to estimate the direction of a Pinger relative to the hydrophone array placed in an autonomous underwater vehicle. If only a line-of-sight path exists between a Pinger and a hydrophone array, the direction of a Pinger can be extracted by measuring the phase delay between hydrophones. However, the actual underwater environment results in multipaths and fading between a Pinger and hydrophone arrays. As a result, the received signal is a sum of multiple copies of the same transmitted signal with different time delays, magnitudes, phases, and angles of arrival. Hence, there is a need for more sophisticated algorithms that can decompose a received signal into signals arrived from different paths. As a result, we consider this problem as the bearing estimation problem in the multipath fading channel.

The problem of bearing estimations have been studied by many researchers [1]-[4]. In [1], the authors proposed the JADE algorithm to jointly estimate both angle of arrival and time delay. The JADE technique is first estimate the channel which requires the training bits, and follows by estimating the parameters of interest. The results are very promising. However, in the problem of Pinger location, it is difficult to obtain any training data since the distance, time delay and bearing are usually unknown. The work by [2] introduced the maximum likelihood to the problem of estimating the angles of arrival. The authors employed the alternating projection that transforms the multivariate nonlinear maximization problem into a sequence of one-dimensional maximization problem to find the solution of the maximum likelihood estimate. The main limitation of this algorithm is that the number of multi-paths must be known in advance. This condition cannot be obtained in most practical situations. Similarly, the work in [3] tried to address this problem by extending the concept in [2] to the problem of joint estimation of angles of arrival and time delays. The authors used the two-stage signal subspace-based processing approach, with or without training data, to estimate the time delays first and then the angles of arrival for each estimated time delay. However, this work did not consider the prior knowledge on the number of multi-paths into consideration.

As a result, our paper attempts to address the problem of unknown number of multi-paths by applying the sparse representation to extract signals arrived from the different paths from each other. The sparse representation of a signal [5] has been proven to be very powerful in acquiring, representing, un-mixing, compressing high-dimensional signals. The sparse representation relies on the fact that signals such audio and images are naturally sparse with respect to some basis set of signal. To apply this idea, we assume that the received signal at a hydrophone is a summation of a line-of-sight signal and few non-line-of-sight multipath signal. The orthogonal matching pursuit (OMP) studied in [6] is employed to decompose the received signal. Next, the subspace technique used in [3] is then employed to estimate the bearing angles of each path.

The rest of the paper is organized as follows. We will introduce the problem statement considered in this paper in the Section II. In Section III, we will propose the main algorithms. The performance of our proposed algorithm will be demonstrated in Section IV on both simulated data and real data. In the last section, we will conclude our work.

II. Problem Statement

Let $s[n]$ be a transmitted signal pulse from a Pinger at time n. A simple example of $s[n]$ is given as

$$s[n] = \begin{cases} a_0 \cos(2\pi f_0 n + \zeta_0) & n = 0, 1, 2, \ldots, N_0 \\ 0 & \text{elsewhere} \end{cases} \quad (1)$$

where a_0, f_0, ζ_0 and N_0 are the amplitude, frequency, phase, and pulse length of a transmitted pulse, respectively. This pulse is transmitted over fading channel through P paths. In general,

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the value of P is unknown. In the noise-free environment, the received complex signal at the i-th hydrophone from the p-th path is given by

$$y_{i,\tau_p}[n] = g_{\tau_p} \times e_i(\theta_{\tau_p}, \phi_{\tau_p}) \times s[n - \tau_p] \tag{2}$$

where g_{τ_p} is the complex gain, $e_i(\theta_{\tau_p}, \phi_{\tau_p}) = e^{jd_i(\theta_{\tau_p}, \phi_{\tau_p})}$ is the phase delay of the i-hydrophone from the arrival elevation and azimuth angles, θ_{τ_p} and ϕ_{τ_p}, $\tau_p \in \{0, 1, \ldots, N-1\}$ is the travel time of an audio signal from a Pinger to the hydrophone array along the p-th path, and $n = 0, \ldots, N-1$. Here, we assume that all paths have distinct time delays, i.e., $\tau_p \neq \tau_q$ for $p \neq q$, and N is the number of collected samples. In this paper, we define

$$d_i(\theta_{\tau_p}, \phi_{\tau_p}) = \Psi f(\theta_{\tau_p}, \phi_{\tau_p}) \tag{3}$$

where $\Psi = 2\pi/\lambda$ is a function of wavelength λ, and $f(\theta_{\tau_p}, \phi_{\tau_p})$ is a function of elevation and azimuth angles. Next, we assume that the line-of-sight (LOS) path between a Pinger and the hydrophone array is a shortest path with the smallest τ_p.

The overall received signal at the i-th hydrophone from all P paths can be written as

$$\begin{aligned} y_i[n] &= \textstyle\sum_{p=0}^{P-1} y_{i,\tau_p}[n] = \sum_{m=0}^{N-1} y_{i,m}[n] \\ &= \textstyle\sum_{m=0}^{N-1} g_m \times e_i(\theta_m, \phi_m) \times s[n-m] \end{aligned} \tag{4}$$

where $|g_m| = 0$ and $e_i(\theta_m, \phi_m) = 1$ if $m \notin \{\tau_p \mid p \in \{0, 1, \ldots, P-1\}\}$, i.e., there is no path with time delay $\tau_p = m$. Eq. (4) can be arranged into a vector format as

$$\mathbf{y}_i = SG\mathbf{e}_i. \tag{5}$$

Here, we define $S = \begin{bmatrix} \mathbf{s}_0 & \cdots & \mathbf{s}_{N-1} \end{bmatrix}$, $G = \mathrm{diag}(\mathbf{g})$, and $\mathbf{e}_i = \begin{bmatrix} e_i(\theta_0, \phi_0) & \cdots & e_i(\theta_{N-1}, \phi_{N-1}) \end{bmatrix}^T$ where $\mathbf{s}_{\tau_p} = \begin{bmatrix} s[0-\tau_p] & \cdots & s[N-1-\tau_p] \end{bmatrix}^T$, $\mathbf{g} = [g_0 \cdots g_{N-1}]^T$ and $\mathrm{diag}(\mathbf{v})$ creates a diagonal matrix from $\mathbf{v}$.

Furthermore, let $Y = \begin{bmatrix} \mathbf{y}_0 & \cdots & \mathbf{y}_{M-1} \end{bmatrix}$ be a collection of observations from all M hydrophones. Hence, we can write

$$Y = SGE = SG \begin{bmatrix} \mathbf{e}_0 & \cdots & \mathbf{e}_{M-1} \end{bmatrix}. \tag{6}$$

Finally, for the noisy environment, the observation matrix is given by

$$Z = Y + W \tag{7}$$

where $Z = [\mathbf{z}_0, \cdots, \mathbf{z}_{M-1}]$, $W = [\mathbf{w}_0, \cdots, \mathbf{w}_{M-1}]$. Note that $\mathbf{w}_m$ is assumed to be a zero-mean white Gaussian noise vector with covariance matrix $\sigma^2 I$.

III. The Optimum Bearing and Time of Arrival Estimation Algorithm

Traditionally, due to the fact that noise is additive white Gaussian, the least square error criteria is used to find the optimum bearing and time of arrivals from all paths, i.e.,

$$(\boldsymbol{\tau}, \boldsymbol{\theta}, \boldsymbol{\phi})^{\mathrm{opt}} = \arg \min_{(\boldsymbol{\tau}, \boldsymbol{\theta}, \boldsymbol{\phi})} \left[\|Z - SGE\|_F^2 \right] \tag{8}$$

where $(\boldsymbol{\tau}, \boldsymbol{\theta}, \boldsymbol{\phi}) = \{(\tau_p, \theta_{\tau_p}, \phi_{\tau_p}) \mid \tau_p = 0, 1, \ldots, N-1\}$, and $\|.\|_F$ denotes the Frobenius norm of a matrix. Clearly, the solution to the above optimization problem yields N sets of time delays and bearings. This can lead to the mislabeling problem of the LOS path between a Pinger and the hydrophone array.

Hence, in this paper, we introduce an additional constraint into Eq. (8) to limit the number of possible paths, i.e., the number of multipaths should be as small as possible. To incorporate this additional constraint, the optimization must be modified. Hence, the optimization problem becomes

$$\begin{aligned} \underset{x}{\text{minimize}} \quad & \|\mathbf{g}\|_0 \\ \text{subject to} \quad & \|Z - SGE\|_F^2 \leq \epsilon \end{aligned} \tag{9}$$

where $\|\mathbf{g}\|_0$ is the number of non-zero entries in $\mathbf{g}$. The problem in Eq. (9) is called the sparse representation problem [5]. In this paper, we employ the orthogonal matching pursuit (OMP) [6] to find the solution in Eq. (9) due to its simplicity. To apply the OMP successfully, we divide the algorithm into two processes, namely arrival time and bearing estimations.

A. Arrival Time Estimation Algorithm

The algorithm for finding the arrival times are given in Algorithm 1. Note that this algorithm first chooses the arrival time, τ_p, with the largest sum of the magnitudes of the correlation coefficients between $s[n - \tau_p]$ and the received pulses $\mathbf{r}_{0,m}[n] = \mathbf{z}_m$ at all hydrophones. After that the OMP projects the received signals onto a subspace orthogonal to $s[n - \tau_p]$ called $\mathbf{r}_{1,m}[n]$ where $\mathbf{r}_{h,m}[n]$, $h = 1, 2, \ldots, h_{max}$, is the signal that is orthogonal to a subspace of all chosen signal pulses corresponding to the estimated arrival times at the iteration number h. The next arrival time with the largest sum of the magnitudes of the correlation coefficients between $s[n - \tau_p']$ and the projected signals is chosen. Then, $\mathbf{r}_{2,m}[n]$ is calculated from the projection of the received signals onto a new orthogonal subspace. This process is repeated until the maximum number of iterations is reached, or the norm of the projected signals less than a predefined value, ϵ'. Let $T = \{\tau_0, \tau_1, \ldots, \tau_{P-1}\}$ be the resulting estimated arrival times and $S_T = S(T) = \left[\mathbf{s}_{\tau_0}, \cdots, \mathbf{s}_{\tau_{P-1}}\right]$ be a matrix containing the corresponding delayed versions of the transmitted signal where P is the number of estimated arrival paths.

B. Bearing Estimation Algorithm

Here, we follow the idea introduced in [3]. From the estimated arrival times in the previous subsection, the received signal can be rewritten as

$$Z = S_T G_T E_T + W \tag{10}$$

where $G_T = \mathrm{diag}([g_{\tau_0}, \cdots, g_{\tau_{P-1}}])$, and

$$E_T = \begin{bmatrix} e_0(\theta_{\tau_0}, \phi_{\tau_0}) & \cdots & e_{M-1}(\theta_{\tau_0}, \phi_{\tau_0}) \\ \vdots & & \\ e_0(\theta_{\tau_{P-1}}, \phi_{\tau_{P-1}}) & \cdots & e_{M-1}(\theta_{\tau_{P-1}}, \phi_{\tau_{P-1}}) \end{bmatrix}.$$

Hence, the optimization problem reduced to

$$(\boldsymbol{\theta}, \boldsymbol{\phi})^{\mathrm{opt}} = \arg \min_{(\boldsymbol{\theta}, \boldsymbol{\phi})} \left[\|Z - S_T G_T E_T\|_F^2 \right] \tag{11}$$

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Algorithm 1 Finding Solution of Eq. (9) by using the orthogonal matching pursuit

1: $h = 0$, TERMINATED = FALSE, $T_0 = \emptyset$, $S_0 = S(T_0) = \emptyset$, and $R_0 = [\mathbf{r}_{0,0}, \mathbf{r}_{0,1}, \cdots, \mathbf{r}_{0,M-1}] = Z$.
2: **for** n=0 to N-1 **do**
3: $\hat{s}_n = \frac{s_n}{\|s_n\|}$
4: **end for**
5: **while** $h < h_{max}$ and not TERMINATED **do**
6: $h + 1 \longrightarrow h$
7: $\hat{n} = \arg\max_n \sum_{m=0}^{M-1} |\langle \mathbf{r}_{h-1,m}, \hat{s}_n \rangle|$.
8: $T_h = T_{h-1} \cup \{\hat{n}\}$ and $S_h = S(T_h)$
9: $Q = S_h \left(S_h^H S_h\right)^{-1} S_h^H$.
10: $R_h = (I - Q)Z$
11: **if** $\|R_h\| < \epsilon'$ **then**
12: TERMINATED = TRUE
13: $T = T_h$, $P = h$, and $S_T = S(T)$.
14: **end if**
15: **end while**

By projecting Eq. (11) onto a space spanned by $\{\mathbf{s}_{\tau_0}, \cdots, \mathbf{s}_{\tau_{P-1}}\}$, and the optimization problem can be reduced to

$$(\boldsymbol{\theta}, \boldsymbol{\phi})^{\text{opt}} = \arg \min_{(\boldsymbol{\theta}, \boldsymbol{\phi})} \left[\|C - G_T E_T\|_F^2\right] \quad (12)$$

where $C = \left[S_T^H S_T\right]^{-1} S_T^H Z$, and $(.)^H$ denotes the matrix Hermitian operation. We note here that C and $G_T E_T$ are $P \times M$ matrices where their p-th rows are associated with the p-th path between a Pinger and the hydrophone array, and m-th columns are associated with the m-th hydrophone. Let $\mathbf{c}_p$ denote the p-th row of C in the column format. We can write the optimization problem for a path with the arrival time τ_p as

$$\left(\theta_{\tau_p}, \phi_{\tau_p}\right)^{\text{opt}} = \arg \min_{(\theta_{\tau_p}, \phi_{\tau_p})} \left[\|\mathbf{c}_p - g_{\tau_p} \mathbf{e}(\theta_{\tau_p}, \phi_{\tau_p})\|_2^2\right] \quad (13)$$

where $\mathbf{e}(\theta_p, \phi_p) = [e_0(\theta_p, \phi_p), \cdots, e_{M-1}(\theta_p, \phi_p)]^T$.

The optimization term in Eq.(13), can be rewritten as

$$\|\mathbf{c}_p - g_{\tau_p} \mathbf{e}(\theta_{\tau_p}, \phi_{\tau_p})\|_2^2 = K - 2\text{Re}(g_p \mathbf{c}_p^H \mathbf{e}(\theta_{\tau_p}, \phi_{\tau_p})) \quad (14)$$

where $K = \|\mathbf{c}_p\|_2^2 + M|g_p|^2$ has a positive real value. The value of g_p can be estimated from $\mathbf{c}_p$ and it is given by

$$g_p = \frac{\mathbf{e}^H(\theta_{\tau_p}, \phi_{\tau_p}) \mathbf{c}_p}{\mathbf{e}^H(\theta_{\tau_p}, \phi_{\tau_p}) \mathbf{e}(\theta_{\tau_p}, \phi_{\tau_p})} = \frac{\mathbf{e}^H(\theta_{\tau_p}, \phi_{\tau_p}) \mathbf{c}_p}{M}. \quad (15)$$

By substituting Eq. (15) into Eq. (14), we yield

$$\|\mathbf{c}_p - g_{\tau_p} e(\theta_{\tau_p}, \phi_{\tau_p})\|_2^2 = \|\mathbf{c}_p\|_2^2 - \frac{1}{M} \left|\mathbf{e}^H(\theta_{\tau_p}, \phi_{\tau_p}) \mathbf{c}_p\right|^2. \quad (16)$$

Hence, the optimization problem becomes

$$\left(\theta_{\tau_p}, \phi_{\tau_p}\right)^{\text{opt}} = \arg \max_{(\theta_{\tau_p}, \phi_{\tau_p})} \left|\mathbf{e}^H(\theta_{\tau_p}, \phi_{\tau_p}) \mathbf{c}_p\right|^2. \quad (17)$$

The above problem can be solved using a traditional optimization technique such as the Newton method [7].

IV. Example

In this section, we examined the performance of the proposed algorithm on both simulated and real datasets. In both cases, we assume that the hydrophone array consists of four hydrophones located at the four corners of a square with the length d on each side. Hence, the hydrophones are located at coordinates $(-d/2, -d/2)$, $(-d/2, d/2)$, $(d/2, d/2)$ and $(d/2, -d/2)$ where $d = 2$ cm. The azimuth and elevation angles (ϕ and θ) are measured from x and z axes, respectively. Fig. 1 displays the hydrophone setup used in this experiment.

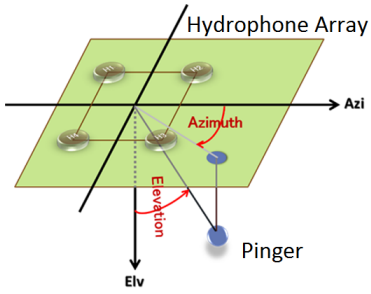


Fig. 1. Hydrophone arrangement considered in this paper

A. simulation

In the simulation, we assume that a Pinger transmits a signal given in Eq.(1) for 4 milliseconds at $f_0 = 25$ kHz. This signal is sampled with an analog to digital converter at the sampling rate of 192 kHz. Hence, we have $N_0 = 768$ samples. The Pinger is located at θ_0 and ϕ_0 with respect to the hydrophone array axes, and the LOS distance between the Pinger and hydrophone array results in the signal pulse arrives at $n = \tau_0$. In this experiment, we assume that θ_0 and ϕ_0 are uniformly distributed from 0 to $\pi/2$ and from $-\pi$ to π, respectively. Next, the amplitude, a_0, is assumed to Rayleigh distributed with the mean of 1 and the phase, ζ_0 is a uniform random variable between $-\pi$ and π. To simulate the multipath environment, we assume that there are other $P - 1$ paths that a signal can take. The p-th reflected signal arrives at the bearing angles θ_p and ϕ_p and the arrival time $\tau_p > \tau_0$ with the amplitude of a_p. Here, θ_p and ϕ_p are again assumed to be uniformly distributed between 0 and $\frac{\pi}{2}$ and and $-\pi$ and π respectively for $p = 0, 1, \ldots, P - 1$. The amplitude of the reflected signal (a_p for $p = 1, \ldots, P - 1$) is also assumed to be Rayleigh distributed with mean $\mu_p < 1$, and the arrival time τ_p for $p > 0$ is assumed to be uniformly distributed between 0 and τ_{max}. The received signal is also corrupted by the additive white Gaussian noise with mean of zero and variance of σ_n^2.

In this experiment, we examined the effect of noise to the system performance. Here, assumed $P = 11$ and varied σ_n^2 form 0.01 to 10, or the line-of-sight (LOS) signals have the signal to noise ratio (SNR) of -10 to 20 dB. τ_0 and τ_{max} are equal to 1 and 6 ms, respectively. The root mean square errors (RMSs) after 100 runs for τ_0, θ_0 and ϕ_0 are different SNRs are given in Tab.I. Here, we observed that, as SNR decreases,

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TABLE I
ROOT MEAN SQUARE ERRORS OF THE ESTIMATED PARAMETERS FOR THE LOS PATH VERSUS SNR

SNR (dB)	Proposed Method			Convolutional Filter		
	τ_0 (μs)	ϕ_0	θ_0	τ_0 (μs)	ϕ_0	θ_0
-10	1199	74.2°	27.0°	638.0	76.4°	24.2°
0	124.9	32.4°	16.0°	564.6	63.75°	22.4°
10	33.0	22.3°	9.84°	577.9	64.8°	25.7°
20	40.6	9.12°	8.95°	589.2	72.1°	24.7°

TABLE II
ROOT MEAN SQUARE ERRORS OF THE ESTIMATED PARAMETERS FOR THE LOS PATH VERSUS NUMBER OF NLSPS

NLSP	Proposed Method			Convolutional Filter		
	τ_0 (μs)	ϕ_0	θ_0	τ_0 (μs)	ϕ_0	θ_0
5	3.72	5.89°	8.26°	337.3	41.2°	19.9°
10	33.0	22.3°	9.84°	589.2	72.1°	24.7°
15	136.3	25.1°	9.7°	770.1	87.3°	29.9°
20	156.2	30.3°	15.6°	799.8	89.2°	29.9°
25	223.4	45.6°	20.5°	859.1	80.5°	28.1°
30	202.0	39.2°	16.5°	851.2	93.7°	30.0°

the estimation errors in terms of arrival times and bearings increases. For the comparison, we employed the matched filter to estimate the arrival times. Here, we associated the peaks of the output of the matched filter to the LOS signal where $\tau_{\text{match}} = t_p - N_0$ is the estimated arrival time of the LOS signal using match filter. Next, the same technique for bearing estimation is used to estimate the azimuth and elevation angles where $S_T = [s_{\tau_{\text{match}}}]$. From Tab. I, we observed that our algorithm outperforms the convolution filter approach for high and medium SNR. However, when SNR is small, our approach and convolution filter approach yield roughly the same performance. This suggests that our algorithm requires sufficiently high SNR to achieve some improvement over the traditional method.

Next, we examined the effect of multipaths to the system performance. Here, we varied the number of non-line-of-sight paths (NLSPs) from 5 to 30 while setting the SNR of the LOS path to 10 dB. The results are given in Tab. II. We observed that the performance of our approach degrades slowly as the number of NLSPs increases, and the performances of the convolutional filter approach are significantly worse than our proposed technique.

B. Real Dataset

In this real data set, we collected the data by using a Pinger model ALP-365 from Teledyne that produced the acoustic pulse every 2 seconds. Each pulse is a constant phase sinusoidal with the frequency of 30 kHz. This acoustic signals are collected using four Reson TC-4013 hydrophones arranged as shown in Fig. 1. An Analog Devices SHARC-21369 digital signal processor is used to collect the data at the sampling rate of 192 kHz.

The Pinger was moved around at five locations (Tab. III) the swimming pool in Kasetsart University while the hydrophone array is kept at a fixed location. Since the arrival times is unknown, we examined the performance of our algorithm in term of bearing angles only. The resulting bearing angles are given in Tab. IV. From our results, it can be observed that our algorithm provides accurate results for small elevation angles.

TABLE III
THE BEARING ANGLES USED IN THE REAL EXPERIMENT

Experiment	1	2	3	4	5
ϕ_0	30°	30°	30°	75°	70°
θ_0	52°	26°	60°	38°	83°

TABLE IV
THE ESTIMATED BEARING ANGLES FROM OUR PROPOSED ALGORITHM

Experiment	1	2	3	4	5
ϕ_0	27.8°	31.8°	24.5°	65.8°	65.2°
θ_0	50.8°	22.7°	47.0°	34.0°	68.1°

V. CONCLUSION

This paper introduced a new Pinger bearing estimation based on the concept of sparse representation and two-stage signal subspace-based processing approach. Here, we decomposed the received signal into the summation of multiple copies of the transmitted signals with different arrival times and bearings. The goal of the sparse representation is to choose the solution as the few number of paths as possible where the desired errors between received signal and reconstructed signal is achieved. From our both simulation and real experiments, we concluded that our proposed technique can successfully estimate the bearing angles of the line-of-sight path.

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